

INDEPENDENT PEER-REVIEW PACKET

EC-007

Planar Coalesced Six-Pin Conditional Jet Law

Cold Review Packet 3 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	3 of 20
Object ID	EC-007
Object class	Exact Gaussian conditional law
Exact statement	Under the planar Bargmann-Fock derivative convention with $P=(f,f_x,f_y,f_{xx},f_{xy},f_{xxx})=(b,0,0,0,0,2)$ and $J=(f_{yy},f_{xxy},f_{xyy},f_{yyy})$, $E[J P]=(-b,0,0,0)$ and $\text{Cov}(J P)=\text{diag}(2,2,2,6)$.
Internal terminal label	CLOSED - EXACT PLANAR COALESCED SIX-PIN CONDITIONAL JET LAW
Terminal date	2026-07-21
Recommended expertise	Gaussian random fields / covariance calculus / linear algebra

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Exact covariance differentiation and Schur complement; multiple independent matrix reconstructions.
Prior review record	Independent exact matrix routes agreed on the mean, ordering, and diagonal covariance.
Binding exclusions	Planar coalesced law only; not exact torus, finite-r, nine-jet, Palm, or adjacency law.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: Planar coalesced law only; not exact torus, finite-r, nine-jet, Palm, or adjacency law.

4. Object-specific hostile checks

- Reconstruct all derivative covariances from the kernel.
- Verify the declared ordering before taking the Schur complement.
- Test whether any off-diagonal entry survives under an alternate derivative-sign convention.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-CLS-PROP-008-v1.0 — PROPOSED TERMINAL CLOSURE OF THE PLANAR COALESCED SIX-PIN CONDITIONAL JET LAW

Artifact ID: GP-CLS-PROP-008-v1.0

Author: OpenAI GPT-5.6 Thinking

Created: 2026-07-21

Class: CLS-PROP — additive terminal-closure proposal

Status: READY FOR HUMAN APPROVAL

Authority: none until Dylan M. Roy accepts or modifies the terminal statement

Canonical impact: NONE before approval

Queue item: EC-007

1. EXACT OBJECT TO BE CLOSED

Let f be the planar Bargmann–Fock Gaussian field with covariance

$$K(x, y) = \exp(-(x^2 + y^2)/2).$$

At one point, define the ordered six-pin vector

$$P = (f, f_x, f_y, f_{xx}, f_{xy}, f_{xxx})$$

and the ordered free-jet vector

$$J = (f_{yy}, f_{xxy}, f_{xyy}, f_{yyy}).$$

Condition on the coalesced pin values

$$P = (b, \theta, \theta, \theta, \theta, 2).$$

Then the exact conditional Gaussian law is

$$E[J \mid P = (b, \theta, \theta, \theta, \theta, 2)] = (-b, \theta, \theta, \theta),$$

$$\text{Cov}(J \mid P) = \text{diag}(2, 2, 2, 6).$$

In particular,

$$E[f_{yy} \mid \text{pins}] = -b.$$

Proposed terminal label:

CLOSED — EXACT PLANAR COALESCED SIX-PIN CONDITIONAL JET LAW

2. FIRST-PRINCIPLES DERIVATION

For derivative multi-indices α and β at the same point,

$$\begin{aligned} \text{Cov}(\partial^\alpha f, \partial^\beta f) \\ = (-1)^{|\beta|} \partial^\alpha (\alpha + \beta) \kappa(\theta). \end{aligned}$$

Using the exact order declared above, the six-pin covariance is

$\Sigma_{PP} =$

$$\begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -3 & 0 & 0 & 0 & 15 \end{bmatrix}.$$

The raw free-jet covariance is

$\Sigma_{JJ} =$

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 3 \\ 0 & 0 & 3 & 0 \\ 0 & 3 & 0 & 15 \end{bmatrix}.$$

The free-jet/pin cross-covariance is

$\Sigma_{JP} =$

$$\begin{bmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 3 \\ 0 & 0 & -3 & 0 & 0 & 0 \end{bmatrix}.$$

Exact matrix elimination gives the regression matrix

$\Sigma_{JP} \Sigma_{PP}^{-1} =$

$$\begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 \end{bmatrix}.$$

Applied to the conditioned pin vector $(b, 0, 0, 0, 0, 2)$, this yields

$$E[J \mid \text{pins}] = (-b, 0, 0, 0).$$

The Schur complement is

$$\Sigma_{JJ} - \Sigma_{JP} \Sigma_{PP}^{-1} \Sigma_{PJ} \\ = \text{diag}(2, 2, 2, 6).$$

No asymptotic argument, simulation, or unproved theorem is used in these identities.

3. EVIDENCE AND INDEPENDENT REVIEW

Primary executable source:

- C047R_verification_battery_01.py, Drive ID 1oUx_WO82T_3xp3jQX9FlbfTW0Sb9apk-, source SHA-256 149920344bd4f0ad7f1b0f89793a229894bf2bb31c853315322a0a097e676c76. The source encodes the exact planar covariance, the stated pin/free-jet order, the Schur complement, and the f_{yy} regression row.

Independent OpenAI re-execution:

- GP-AUD-011-v1.0, Drive ID 1i0jWKL1-hhe2yIQhlzQeK4MR51vXbLmpfrL2dYqUYgk, downloaded and hashed the exact source, executed it unmodified under SymPy 1.14.0, and separately reconstructed the covariance and regression calculation. It records both results as CONFIRMED-BY-REEXECUTION.

Independent C047R acceptance:

- CW-RSP-012-v1.0, Drive ID 1q10Wg4YNj_5cF8a7kOPJhAtJJXzM9_Qo, accepts GP-AUD-011 in full and explicitly confirms the covariance $\text{diag}(2,2,2,6)$ and regression $E[f_{yy}|\text{pins}]=-b$.

Object-basis crosswalk:

- CW-DER-013-v1.0, Drive ID 15JfjUCQZFRQni69JYfzbRQFQ2i3vgMlp, identifies the free cubic-order jets as $(f_{yy}, f_{xxy}, f_{xyy}, f_{yyy}) = (q, a, w, z)$, consistent with the object closed here.

Fresh OpenAI exact reconstruction:

- This closure proposal independently rebuilds the derivative covariance matrices, regression matrix, and Schur complement directly from $K(x,y) = \exp(-(x^2+y^2)/2)$.

4. LOAD-BEARING SCOPE LIMIT

This closure applies only to the declared PLANAR_COALESCED Bargmann–Fock object and the exact jet ordering

$$P = (f, f_x, f_y, f_{xx}, f_{xy}, f_{xxx}), \\ J = (f_{yy}, f_{xxy}, f_{xyy}, f_{yyy}).$$

It does NOT assert literal equality for the exact normalized-periodized finite-r torus field. The exact-torus conditioned law converges to this planar coalesced law but carries exponentially small periodization corrections and finite-r corrections. Future citations must therefore distinguish:

- PLANAR_COALESCED_SIX_PIN_JET_LAW — closed by this package;
- EXACT_TORUS_COALESCED_JET_LAW — a distinct fixed-torus object;
- EXACT_TORUS_FINITE_R_JET_LAW — a distinct finite-r object.

5. DEPENDENCIES AFFECTED

After approval, this closure permits the exact planar conditional mean and covariance to be cited without reopening the Schur-complement calculation whenever the declared object and ordering match exactly.

It supports:

- • planar local-model sampling and symbolic checks;
- • audit of limiting nine-jet and free-jet calculations;
- • object-card validation distinguishing planar coalesced from exact-torus and finite-r laws;
- • future collision-map covariance derivations that explicitly use the free-jet covariance $\text{diag}(2,2,2,6)$.

6. DEPENDENCIES UNAFFECTED

This closure does not:

- • prove the exact finite-r or fixed-torus covariance equals the planar matrix;
- • prove adjacency positivity, adjacency-to-one, branch capture, or P0.2;
- • verify Palm normalization, determinant weighting, or probability estimates;
- • change Theorem B or the live q_0 theorem state;
- • promote any machine root or successor machine state;
- • validate every assertion in the original battery;
- • rehabilitate the battery's rejected weak $\text{abs}(\text{residual}) < 1$ geometry control;
- • alter any historical source file.

7. TERMINAL RECORD AND REGISTER ACTIONS AFTER APPROVAL

After Dylan M. Roy approves or modifies the terminal statement:

1. Move EC-007 from Easy Closure Queue to Closure Log.

2. Record terminal label:

CLOSED — EXACT PLANAR COALESCED SIX-PIN CONDITIONAL JET LAW

3. Register object tags:

PLANAR_COALESCED**SIX_PIN_CONDITIONING****FREE_JETS_Q_A_W_Z****CONDITIONAL_MEAN_MINUS_B****CONDITIONAL_COVARIANCE_DIAG_2_2_2_6****NOT_EXACT_TORUS_FINITE_R****4. Cross-link GP-AUD-011 and CW-RSP-012.****5. Remove this identity from active re-derivation and routine review queues.****6. Preserve every source, audit, response, and correction artifact unchanged.****8. HUMAN APPROVAL REQUIREMENT**

This item is mathematically closure-ready but remains a human terminal decision under OP-PROT-004.

Proposed approval statement:

I approve GP-CLS-PROP-008-v1.0. Close the exact planar coalesced six-pin conditional law for $P=(f, f_x, f_y, f_{xx}, f_{xy}, f_{xxx})=(b, 0, 0, 0, 0, 2)$ and $J=(f_{yy}, f_{xxy}, f_{xyy}, f_{yyy})$, with $E[J|P]=(-b, 0, 0, 0)$ and $\text{Cov}(J|P)=\text{diag}(2, 2, 2, 6)$. Preserve the distinction from exact-torus coalesced and exact-torus finite-r conditioned laws.

9. CORRECTION PATH

If a later audit finds an algebraic, ordering, sign, or object-definition error, do not edit this document silently. File an additive erratum identifying the exact failed line, affected dependencies, and corrected terminal status.

END OF PROPOSED CLOSURE PACKAGE

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT PLANAR COALESCED SIX-PIN CONDITIONAL JET LAW
Closure date	2026-07-21
Evidence basis	Exact covariance differentiation and Schur complement; multiple independent matrix reconstructions.
Independent review already recorded	Independent exact matrix routes agreed on the mean, ordering, and diagonal covariance.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- Planar coalesced law only; not exact torus, finite-r, nine-jet, Palm, or adjacency law.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date _____

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: <https://docs.google.com/document/d/1kBBCBiYJ5eQ0pNduToaYTIJxA-7vQOmMSzCABt3ABtk/edit>

Closure / terminal record: https://docs.google.com/document/d/1Zm65j9BZe_z3P_8qP7rfZukRU-IWUE8h8bDqXdDsIQ/edit

Local source snapshot used to construct this packet: GP-CLS-PROP-008-v1.0 — Proposed terminal closure of the planar coalesced six-pin conditional jet law

Local source snapshot SHA-256 (DOCX export, transport checksum only):
a3f8c44f4659c956f58a037273490208c6175503a8fac1249ec13beab2e46a23

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.